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Q. 1. The surface of equation $x^2 + 2y^2 + 3z^2 = 1$ is called

- (A) a sphere (B) a paraboloid
(C) an ellipsoid (D) a plane

Q. 2. The surface of equation $z = -5 + \sqrt{x^2 + y^2}$ is called

- (A) a sphere (B) a paraboloid
(C) an ellipsoid (D) a cone

Q. 3. Let

$$f(x, y) = \frac{e^{x^3-y^8} \sin^9(x^3 - y^8)}{\ln(1 + \cos^{10}(x^3 - y^8)) + x^3 - y^8}.$$

Evaluate $f_x(1, -1)$.

- (A) 0 (B) -1
(C) 1 (D) ≈ 1.2341

Q. 4. Let

$$f(x, y) = x^6(y^4 + 1) + \frac{(x+1)^2}{x^2 + y^4 + 1} [\sin(x + y^2) + x^8 \ln(\cos^2(x^8 + y^9))].$$

Evaluate $f_{xy}(-1, 2)$.

- (A) -192 (B) 32
(C) -121 (D) -230

Q. 5. Let

$$g(t, s) = f(3t^2 - 4s, 5ts),$$

where f is a function of two variables. What is the formula for g_t ?

- (A) $g_t(t, s) = f_x(3t^2 - 4s, 5ts) \cdot 6t + f_y(3t^2 - 4s, 5ts) \cdot 5s$
(B) $g_t(t, s) = f_x(3t^2 - 4s, 5ts) \cdot 6t + f_y(3t^2 - 4s, 5ts) \cdot 5t$
(C) $g_t(t, s) = f_x(3t^2 - 4s, 5ts) \cdot (-4) + f_y(3t^2 - 4s, 5ts) \cdot 5s$
(D) $g_t(t, s) = f_x(3t^2 - 4s, 5ts) \cdot (-4) + f_y(3t^2 - 4s, 5ts) \cdot 5t$

Q. 6. Let $g(t, s)$ be a function of two variables t and s . Given that $g_t(11, 0) = -1$. Let f be the function of three variables defined by

$$f(x, y, z) = g(x^2 - y + 3z, xyz).$$

Evaluate $f_y(3, -2, 0)$.

- (A) 1 (B) -1
(C) -11 (D) 9

Q. 7. Use the linear approximation to find an approximate value for $f(-1.99, 3.01)$, given that

$$f(-2, 3) = -2, \quad f_x(-2, 3) = -1, \quad f_y(-2, 3) = -3.$$

- (A) -2.04 (B) -2.13
(C) -1.99 (D) -1.97

Q. 8. Use the linear approximation to find an approximate value for $f(4.01, 3.02)$, given that

$$f(4, 3) = -4, \quad f_x(4, 3) = 5, \quad f_y(4, 3) = -2.$$

- (A) -4.04
(C) -4.01

- (B) -3.97
(D) -3.99

Q. 9. An equation of the tangent plane to the surface of equation $z - 3x + 4y + 1 = 0$ at the point $M(1, 0, 2)$ is

- (A) $3x + 4y - z - 1 = 0$
(C) $x - 4y - 5z + 9 = 0$

- (B) $3x - 4y - z - 1 = 0$
(D) $x + 4y - 5z + 9 = 0$

Q. 10. Given that the function

$$f(x, y) = 12 + 4x + 2x^2 - 6y - 2xy + y^2$$

has a local minimum. Find a point M where this minimum attains.

- (A) $M(1, 4)$
(C) $M(1, -4)$

- (B) $M(-1, 4)$
(D) $M(1, 3)$

Q. 11. Given that the function

$$f(x, y) = 5 + 30x - 6x^2 + 3x^3 - 18xy + 3xy^2$$

has a local minimum. Find a point M where this minimum attains.

- (A) $M(1/3, 3)$
(C) $M(1, 3)$

- (B) $M(-1/3, 3)$
(D) $M(-1, 1/3)$

Q. 12. How many local extreme points does the function

$$f(x, y) = 1 + x^3 - 2xy - 3xy^2$$

have?

- (A) 0
(C) 2

- (B) 1
(D) 3

Q. 13. Find the minimum value of the function

$$f(x, y, z) = \frac{(x-1)^2 + (y-2)^2 + (z-3)^2}{4}$$

on the sphere $\{(x, y, z) : x^2 + y^2 + z^2 = 9\}$.

- (A) 0.15
(C) $(30 - 8\sqrt{14})/4$

- (B) 0.25
(D) $(23 - 6\sqrt{14})/4$

Q. 14. Evaluate the mass of the lamina that occupies the rectangle

$$\{(x, y) : -1 \leq x \leq 3, 2 \leq y \leq 4\}$$

and has the density function $\rho(x, y) = x^2 + y^2$.

- (A) $17/3$
(C) $280/3$

- (B) $35/3$
(D) $158/3$

Q. 15. Evaluate the mass of the lamina that occupies the rectangle

$$\{(x, y) : -1 \leq x \leq 3, 2 \leq y \leq 5\}$$

and has the density function $\rho(x, y) = x^2 y^2$.

- (A) 390
(C) 180

- (B) 350
(D) 364

Q. 16. Evaluate the mass of the lamina that occupies the disk

$$\{(x, y) : x^2 + y^2 \leq 9\}$$

and has the density function $\rho(x, y) = x^2 + y^2 + 1$.

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|-----------------|---------------|
| (A) 121 | (B) 121π |
| (C) $6723\pi/2$ | (D) $99\pi/2$ |

Q. 17. Evaluate

$$\iint_D 1dA,$$

where D is the region bounded inside the circle $x^2 + y^2 = 4$ and outside the square of vertices $A(0, 2)$, $B(2, 0)$, $C(0, -2)$, $D(-2, 0)$.

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|----------------|---------------|
| (A) 2π | (B) $\pi + 1$ |
| (C) $4\pi - 8$ | (D) 8π |

Q. 18. Evaluate

$$\iint_D x dA,$$

where D is the region bounded inside the circle $x^2 + y^2 = 9$ and below the line $y = x$.

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|-----------------|-----------------|
| (A) $2\pi - 3$ | (B) 10 |
| (C) $9\sqrt{2}$ | (D) $5\sqrt{3}$ |

Q. 19. Evaluate

$$\iint_D 1dA,$$

where D is the region bounded by the curve $y = \sqrt{x}$ and two lines: $x = 1$ and $x + y = 0$.

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|-----------------|-----------|
| (A) $7/6$ | (B) $4/3$ |
| (C) $3\sqrt{2}$ | (D) $1/3$ |

Q. 20. Let (S) be the part of the paraboloid $z = 2x^2 + 2y^2$ that lies inside the cylinder $x^2 + y^2 = 1$. The projection of (S) onto the plane Oxy is

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|-----------------------------------|---------------------------------|
| (A) the disk $x^2 + y^2 \leq 4$ | (B) the disk $x^2 + y^2 \leq 2$ |
| (C) the disk $x^2 + y^2 \leq 1/2$ | (D) the disk $x^2 + y^2 \leq 1$ |

Q. 21. Let (S) be the part of the paraboloid $z = x^2 + y^2$ that lies below the plane $z = 4x + 6y - 12$. The projection of (S) onto the plane Oxy is

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|---------------------------------|------------------------|
| (A) a disk of radius 2 | (B) a disk of radius 1 |
| (C) a disk of radius $\sqrt{2}$ | (D) a disk of radius 3 |

Q. 22. Let E be the solid bounded inside two spheres $x^2 + y^2 + z^2 = 25$ and $x^2 + y^2 + (z - 5)^2 = 25$. The projection of E onto the plane Oxy is a disk. Evaluate the radius of the disk.

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| (A) $3\sqrt{5}$ | (B) $\sqrt{3}/2$ |
| (C) $5\sqrt{3}/2$ | (D) $3\sqrt{5}/2$ |

Q. 23. Let (P) be the tangent plane to the paraboloid $z = 3x^2 + y^2 - 4$ at the point $M(1, 2, 3)$. Let (C) be the intersection of (P) and the paraboloid $z = x^2 + y^2$. The projection of (C) onto the plane Oxy is a circle. Evaluate its radius.

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| (A) $\sqrt{2}$ | (B) 2 |
| (C) 1 | (D) $\sqrt{3}$ |

Q. 24. Let (S) be the surface of equation $z = x^2 + 2y^2 + 1$. The unit normal vector $\vec{n}$ to (S) at $M(1, -1, 4)$ that goes upward is

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|---|---|
| (A) $\vec{n}(-1/\sqrt{11}, 3/\sqrt{11}, 1/\sqrt{11})$ | (B) $\vec{n}(-2/\sqrt{21}, 4/\sqrt{21}, 1/\sqrt{21})$ |
| (C) $\vec{n}(2/\sqrt{6}, 1/\sqrt{6}, 1/\sqrt{6})$ | (D) $\vec{n}(0, 1/\sqrt{2}, 1/\sqrt{2})$ |

Answer Key for Exam A

Q. 1. (C)	Q. 5. (A)	Q. 9. (B)	Q. 13. (D)	Q. 17. (C)	Q. 21. (B)
Q. 2. (D)	Q. 6. (A)	Q. 10. (A)	Q. 14. (C)	Q. 18. (C)	Q. 22. (C)
Q. 3. (A)	Q. 7. (A)	Q. 11. (C)	Q. 15. (D)	Q. 19. (A)	Q. 23. (A)
Q. 4. (A)	Q. 8. (D)	Q. 12. (A)	Q. 16. (D)	Q. 20. (D)	Q. 24. (B)